

# Calibration of Probabilities

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# 1 Introduction

We frame Supervised Classification as a risk minimization problem because given a labeled training set, a model is trained to minimize a surrogate loss function and the quality of the performance is determined through metrics like classification accuracy. The output of a probabilistic classifier (a number between 0 and 1) is usually interpreted as directly as the probability that a given instance belongs to the positive class; however minimizing the loss function does not guarantee that the interpretation is correct.

A classifier is said to be well calibrated if, among all instances assigned a predicted probability of  $p$ , approximately a fraction  $p$  of them actually belong to the positive class. This property is different from classification accuracy: a model can correctly classify the majority of instances while still producing probability estimates that are too high or too low. This distinction is particularly important because there are applications where the predicted probability is an information itself such as a clinical risk assessment.

Niculescu-Mizil and Caruana showed that several widely used learning algorithms, including boosted trees and random forests, tend to produce poorly calibrated probabilities despite achieving strong discriminative performance. they proposed to use Platt scaling and isotonic regression to correct the miscalibration.

This project uses logistic regression and random forest to evaluate the effectiveness of Platt scaling and isotonic regression. The analysis is performed on two different real word datasets called Heart Disease and Student Dropout.

## 2 Dataset

I used two real world binary classification datasets selected from the UCI Machine Learning Repository. The two datasets differ substantially in size, domain, and feature composition, which allows the analysis of calibration to be carried out under different data conditions.

### 2.1 Heart Disease

The Heart disease dataset contains clinical attributes collected to predict the presence of heart disease in a patient. In the original repository there were four different datasets: `processed.cleveland.data`; `processed.hungarian.data`; `processed.switzerland.data`; `processed.va.data`; in order to obtain a larger and more robust sample I combined this four datasets together. Each instance is described by 13 clinical features, significantly: demographic information (age, sex); measurements taken at rest (resting blood pressure, serum cholesterol, resting electrocardiographic results) and during a stress test (maximum heart rate achieved, exercise-induced angina, ST depression). The original target variable `num` encodes the severity of heart disease on a scale from 0 (absence) to 4; for this project I converted this in a binary variable distinguishing between absence (`num = 0`) and presence (`num > 0`) of heart disease. The three features "slope", "ca", "thal" presented a lot of missing values, concentrated almost entirely in the Hungarian, Switzerland, and V.A. subsets, so I have decided that removing the three features was the best thing to do to avoid substantial bias or noise. After preprocessing, the dataset consists of 920 instances, 10 features, and a reasonably balanced target distribution (411 negative and 509 positive instances).

### 2.2 Student Dropout

The Student Dropout dataset contains various information collected from students in various undergraduate programs, with the goal of predicting academic outcomes. The original target variable classifies each student into one of three categories: "Dropout", "Enrolled", or "Graduate". Since for this project I need a binary classification I changed the target variable: the goal was to distinguish students who dropped out and students that didn't so I merged "Enrolled"

and "Graduate" into a single class. The dataset contains no missing values. After binarization, the class distribution is moderately imbalanced, with 3003 negative and 1421 positive instances.

### 2.3 Dataset confrontation

The Table 1 summarizes the main characteristics of the two datasets. Heart disease is smaller and mildly imbalanced while Student Dropout is almost five times larger and more imbalanced. This difference is important for the calibration step because the size of the calibration set affects the stability of the calibration methods considered.

Table 1: Summary comparison between the two datasets used in this project.

| Dataset         | # Instances | # Features | Classes (0 / 1) |
|-----------------|-------------|------------|-----------------|
| Heart Disease   | 920         | 10         | 411 / 509       |
| Student Dropout | 4 424       | 36         | 3 003 / 1 421   |

## 3 Methodology

### 3.1 Data Split

Each dataset is partitioned in 3 subsets: a training set used to fit classifiers; a calibration set used to fit the post-hoc calibration methods and a test set used to evaluate both the calibrated and uncalibrated models. It is important to calibrate using a set with data that was not used to train to avoid over optimistic calibration estimates and a separate test set is important to obtain unbiased evaluations. The split was performed with proportions of 60% (training), 20% (calibration), and 20% (test) and was stratified by class in order to preserve the original class proportions. Following the split, all features were standardized to zero mean and unit variance.

### 3.2 Classification Models

#### 3.2.1 Logistic Regression

Logistic Regression models the probability that an instance  $\mathbf{x}$  belongs to the positive class as

$$p(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}}, \quad (1)$$

where  $\mathbf{w}$  and  $b$  are the model parameters. The parameters are estimated by minimizing the binary cross-entropy (log-loss) between the predicted probabilities and the true labels,

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{n} \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)], \quad (2)$$

via batch gradient descent. At each iteration, the parameters are updated in the direction opposite to the gradient of the loss,

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \cdot \frac{1}{n} \mathbf{X}^\top (\mathbf{p} - \mathbf{y}), \quad b \leftarrow b - \eta \cdot \frac{1}{n} \sum_{i=1}^n (p_i - y_i), \quad (3)$$

where  $\eta$  denotes the learning rate. The model was trained with  $\eta = 0.1$  for 1 000 iterations, with weights initialized to zero. Convergence was monitored by tracking the training loss across iterations.

### 3.2.2 Random Forest

A random forest is a group of decision trees each trained on a different subset of the original dataset and then the predictions are aggregate to reduce variance compared to a single tree.

Each decision tree is grown recursively: at every node, the algorithm selects the feature and threshold that minimize the weighted Gini impurity of the resulting child nodes, where the Gini impurity of a node is defined as

$$\text{Gini}(\text{node}) = 1 - \sum_c p_c^2 \quad (4)$$

with  $p_c$  denoting the proportion of class  $c$  among the instances in the node. A node becomes a leaf when it reaches the maximum depth or when the number of instances is below a threshold. Randomness is introduced in two ways: first, each tree is trained on a bootstrap sample of the training set, obtained by sampling  $n$  instances with replacement, so that each tree observes a slightly different subset of the data; Second, at each split, only a random subset of  $\sqrt{d}$  features (where  $d$  is the total number of features) is considered as candidates, rather than the full feature set; this decorrelates the individual trees and prevents them from repeatedly selecting the same dominant features. The final probability is obtained averaging all the probabilities predicted by the trees in the forest. The forest used in this project consists of 50 trees, each with a maximum depth of 10 and a minimum of 5 instances required to split a node.

## 3.3 Calibration methods

Both calibration method I used take as input the raw probabilities produced by a trained classifier and learn a mapping from raw to calibrated probabilities. This mapping is subsequently applied to the raw probabilities produced by the same classifier on the test set.

### 3.3.1 Platt Scaling

Platt scaling assumes that the relationship between the raw classifier output and the true class probability can be modeled by a sigmoid function with two parameters  $A$  and  $B$ ,

$$p_{\text{cal}} = \sigma(A \cdot p_{\text{raw}} + B). \quad (5)$$

The parameters  $A$  and  $B$  are estimated by fitting a one-dimensional logistic regression, using the raw probabilities on the calibration set as the single input feature and the true labels as targets.

### 3.3.2 Isotonic Regression

Isotonic regression is a non-parametric calibration method that learns a monotonically non-decreasing step function mapping raw probabilities to calibrated probabilities, without assuming any specific functional form. It was implemented using the Pool-Adjacent-Violators (PAV) algorithm: the instances are first sorted by raw probability, the algorithm then scans the sequence of labels and whenever there is a violation of monotonicity the two blocks are merged in a single block whose value is a weighted average of the original two. This process is repeated until all the sequence is non decreasing

Isotonic regression is considered more flexible compared to Platt scaling as it does not constrain the calibration map to a specific parametric family but this flexibility requires a higher number of effective degrees of freedom so this method is more prone to overfitting

### 3.4 Evaluation metrics

Both calibrated and uncalibrated metrics were evaluated on the test set using the following metrics.

**Accuracy** measures the fraction of correctly classified instances, using a decision threshold of 0.5 on the predicted probability.

**Log-loss** (binary cross-entropy) penalizes confident but incorrect predictions more heavily than accuracy does, and is particularly sensitive to poorly calibrated probabilities near 0 or 1:

$$\text{Log-loss} = -\frac{1}{n} \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)]. \quad (6)$$

**Brier score** is the mean squared error between predicted probabilities and true labels.

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2, \quad (7)$$

**Reliability diagrams** provide a visual assessment of calibration. Predicted probabilities on the test set are grouped into  $B = 10$  equally spaced bins over  $[0, 1]$ ; for each bin, the observed frequency of the positive class among the instances falling into that bin is computed and plotted against the bin’s center. A perfectly calibrated model produces points lying on the diagonal  $y = x$ .

## 4 Results

### 4.1 Baseline pre-calibration

Table 2 reports the test-set performance of both classifiers on both datasets, prior to any calibration. On Heart Disease logistic regression is slightly outperformed by random forest in all three metrics; in student dropout logistic regression has better scores in all three metrics instead. Overall, both classifiers achieve substantially higher accuracy on Student Dropout (86–88%) than on Heart Disease (77–78%), consistent with the larger training set and the more clear-cut nature of the dropout prediction task.

Table 2: Baseline metrics (no calibration) on the test set.

| Dataset         | Model               | Accuracy | Log-loss | Brier |
|-----------------|---------------------|----------|----------|-------|
| Heart Disease   | Logistic Regression | 0.774    | 0.514    | 0.162 |
| Heart Disease   | Random Forest       | 0.780    | 0.475    | 0.154 |
| Student Dropout | Logistic Regression | 0.879    | 0.329    | 0.096 |
| Student Dropout | Random Forest       | 0.868    | 0.335    | 0.101 |

### 4.2 Calibration effect

Table 3 compares the raw predictions to those obtained after applying Platt scaling and isotonic regression. Contrary to expectations the calibration results in an increase of the log-loss in almost every configuration, with isotonic regression producing the largest degradation across the board. Platt scaling yields a small improvement in log-loss and Brier score only for Logistic Regression on Heart Disease (from 0.514 to 0.497, and from 0.162 to 0.160, respectively); in all other cases, both calibration methods worsen or leave essentially unchanged the metrics compared to the uncalibrated baseline. Isotonic regression in particular has the worst results applied to Logistic Regression on Heart Disease, where log-loss increases from 0.514 to 0.904.

Table 3: Comparison of metrics before and after calibration.

| Dataset – Model      | Method   | Accuracy | Log-loss | Brier |
|----------------------|----------|----------|----------|-------|
| Heart Disease – LR   | Raw      | 0.774    | 0.514    | 0.162 |
|                      | Platt    | 0.774    | 0.497    | 0.160 |
|                      | Isotonic | 0.785    | 0.904    | 0.172 |
| Heart Disease – RF   | Raw      | 0.780    | 0.475    | 0.154 |
|                      | Platt    | 0.790    | 0.486    | 0.157 |
|                      | Isotonic | 0.796    | 0.502    | 0.162 |
| Student Dropout – LR | Raw      | 0.879    | 0.329    | 0.096 |
|                      | Platt    | 0.876    | 0.339    | 0.098 |
|                      | Isotonic | 0.865    | 0.466    | 0.097 |
| Student Dropout – RF | Raw      | 0.868    | 0.335    | 0.101 |
|                      | Platt    | 0.869    | 0.354    | 0.104 |
|                      | Isotonic | 0.865    | 0.438    | 0.100 |

### 4.3 Reliability diagram

Figures 1 and 2 show the reliability diagrams for all four model–dataset combinations, with marker size proportional to the number of test instances falling into each probability bin. On Heart Disease (figure 1) the isotonic regression exhibits non monotonic deviation, most notably a drop to 0 in the middle of the logistic regression diagram and a sharp rise to 1 followed by a drop in the random forest. In both cases these deviations correspond to visibly small markers, indicating that very few calibration instances informed those regions of the isotonic map. In contrast bin with a lot of instances show all three curves closely tracking the diagonal.

In the Student Dropout dataset (where the calibration set is almost five time larger) the reliability curves are smoother for all three methods with no strange spikes.

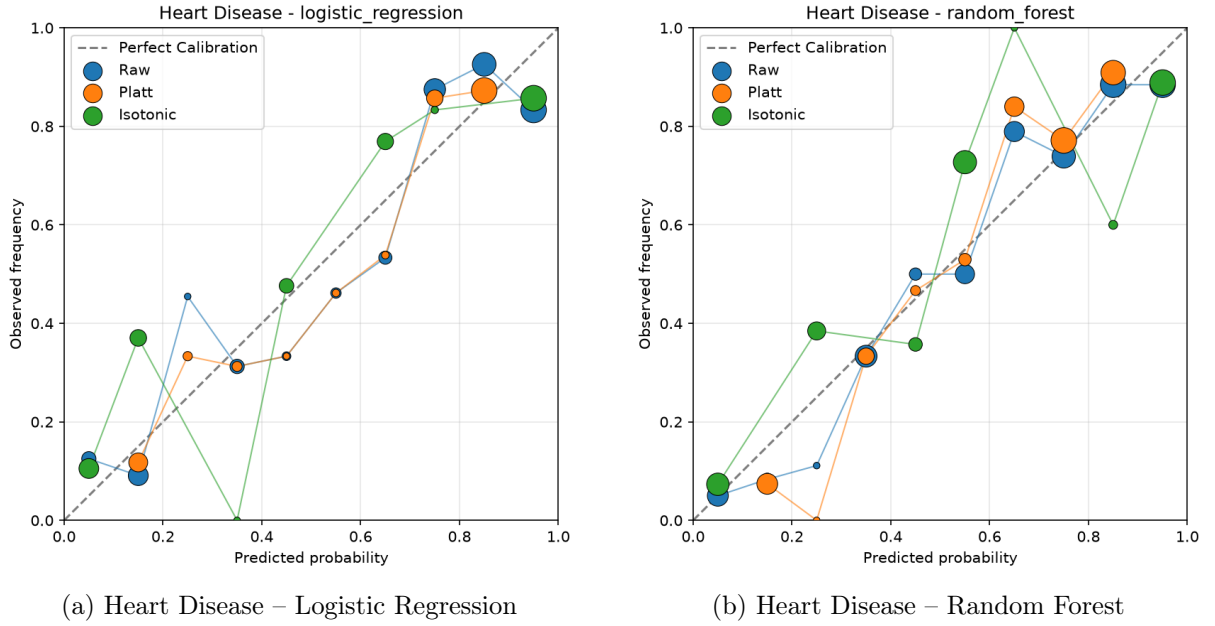


Figure 1: Reliability diagrams for the Heart Disease dataset. Marker size is proportional to the number of test instances in the corresponding bin.

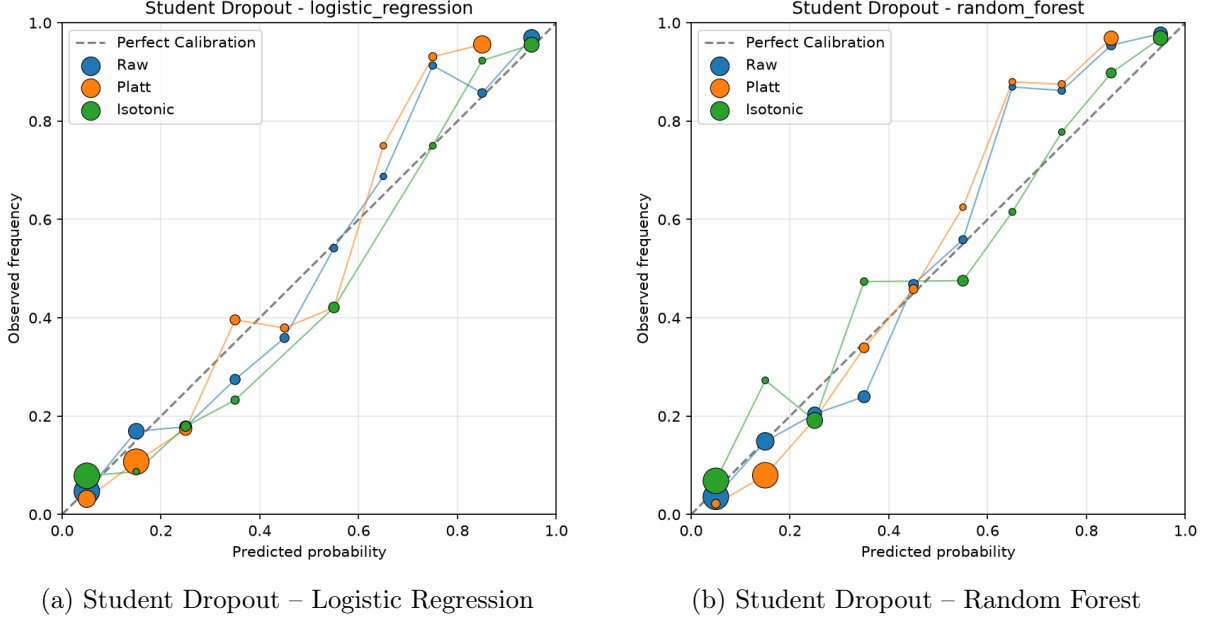


Figure 2: Reliability diagrams for the Student Dropout dataset.

## 5 Discussion

The results showed in section 4 seems to contradict the purpose of calibration, in almost every model-dataset combination both Platt scaling and Isotonic regression failed to improve the evaluation metrics.

The most consistent finding is that isotonic regression performs worse of both raw and Platt scaled probabilities in all four configurations. We can also note that the degradation is far worse in the Heart disease dataset applications. The reason is that isotonic regression is flexible, it has not a fixed shape like Platt scaling so it fits the calibration data as closely as possible. This is a problem for small datasets like heart disease; with so few examples, some probability bins end up containing just a handful of instances, and by chance these instances may all belong to the same class. Isotonic regression then assigns an extreme probability (close to 0 or close to 1) to that bin, even though this does not reflect the real, underlying probability. This results in error during the test set application. This effect exists in Student Dropout too but it’s smaller since the dataset is larger.

Platt scaling is more robust to small calibration because it has only two parameters and this is reflected in the results: Platt scaling does not show the extreme degradation presented by isotonic regression. However, in most other configurations Platt scaling also fails to improve the metrics, suggesting that the limited benefit is not solely a matter of overfitting, but also relates to how well-calibrated the underlying classifiers already are.

On the positive side, the reliability diagrams demonstrate that the loss of aggregate performance is not evenly distributed: the vast majority of probability bins show close agreement between the raw, Platt-scaled, and isotonic curves; this indicates that both models are not mis-calibrated in the beginning. The degradation is only present in a small number of bin and not in every bin; this indicates that it is not a systematic failure of the calibration methods. But, on the negative side, it’s also clear that calibration is very limited in small dataset cases, and sometimes it can even become harmful.

There are a few limitations in this project: the hyperparameter of both models were fixed and not tuned, so different choice can give different results. The calibration set used, especially for Heart Disease, is quite small.

## 6 Conclusions

In this project I implemented two classifiers (Logistic Regression and Random Forest) and two calibration methods (Platt scaling and isotonic regression) and evaluated their calibration properties on two real-world datasets of different size and domain.

Contrary to the expectations the results show that both calibration methods often fail to improve the results; isotonic regression being particularly problematic. The reliability diagrams indicate that this outcome is closely tied to the size of the calibration set: isotonic regression overfits when used on a small number of instances producing extreme and unreliable probability estimates whereas Platt scaling, constrained by its two-parameter functional form, remains comparatively stable even in this setting. On the larger of the two datasets, both methods behave more consistently, though still without a clear net improvement over the uncalibrated probabilities produced by the base classifiers.

These findings are consistent with the general expectations: the flexibility of a calibration method must be matched to the amount of data available to fit it. A flexible method such as isotonic regression needs a large enough dataset to be useful; when this condition is not met, a simpler, more constrained method such as Platt scaling or no calibration at all may be preferable.

## References

- [1] A. Niculescu-Mizil and R. Caruana. *Predicting Good Probabilities with Supervised Learning*. Proceedings of the 22nd International Conference on Machine Learning (ICML), 2005.
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